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# Bihar Engineering University, Patna

## B.Tech 2nd Semester Examination - 2026

***Time: 03 Hours***

***Code: 100201*** ***Subject: Engineering Physics*** ***Full Marks: 70***

***Instructions:-***

*(i) The marks are indicated in the right-hand margin.*
*(ii) There are NINE questions in this paper.*
*(iii) Attempt FIVE questions in all.*
*(iv) Question No. 1 is compulsory.*

***Q.1 Choose the correct option / answer the following (Any seven question only):*** **[2×7=14]**

(a) A glass window may be broken by a distant explosion due to _____
- (i) Interference
- (ii) Resonance
- (iii) Diffraction
- (iv) Polarization

(b) Displacement and velocity graph of SHM is _____
- (i) Parabolic
- (ii) Circular
- (iii) Elliptical
- (iv) Linear

(c) For which angle of scattering is the Compton shift maximum?
- (i) 90°
- (ii) 180°
- (iii) 0°
- (iv) None of the above

(d) Which characterizes a photon?
- (i) Both energy and momentum
- (ii) Momentum, but not energy
- (iii) Energy, but not momentum
- (iv) Neither energy nor momentum

(e) Write the name of the particles produced in pair production?
- (i) Electron and proton
- (ii) Positron and electron
- (iii) Proton and neutron
- (iv) Neutrino and neutron

(f) Ruby LASER is a
- (i) 3 level gas laser
- (ii) 4 level solid state laser
- (iii) Semiconductor laser
- (iv) 3 level solid state laser

(g) Gauss divergence theorem converts _____
- (i) Line integral to volume integral
- (ii) Surface integral to volume integral
- (iii) Surface integral to line integral
- (iv) Surface integral to volume integral & vice versa

(h) Which of the following laws do **not** form a Maxwell equation?
- (i) Gauss's Law of electrostatics
- (ii) Faraday's law of electromagnetic induction
- (iii) Ampere's circuital Law
- (iv) None of the above

(i) The relationship describing the angle at which a beam of X-rays of a particular wavelength diffracts from a crystalline surface is _____
- (i) Snell's law
- (ii) Fresnel's law
- (iii) Bragg's law
- (iv) None of the above

(j) In single slit diffraction the ratio of intensities of first secondary maximum to the central maximum is _____
- (i) 1/28
- (ii) 1/22
- (iii) 1/121
- (iv) 1/5

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***Q.2***

(a) Discuss the structure and working of He-Ne LASER. Draw its energy level diagram. [10]

(b) Explain Coriolis effect and its applications in weather system. [4]

***Q.3***

(a) What is damping? Derive the differential equation for Damped Harmonic Oscillator. Give its solution. [10]

(b) Differentiate between under damped, over damped and critically damped harmonic oscillation including graphical representation. [4]

***Q.4***

(a) What is a diffraction grating? Derive the expression for resultant intensity. Discuss the diffraction pattern due to a grating with intensities distribution curve. [10]

(b) Write a short note on Michelson's interferometer. [4]

***Q.5***

(a) Discuss Kronig-Penney model for the motion of an electron in a periodic potential and the existence of allowed and forbidden energy bands. [10]

(b) State and explain Bloch's theorem for a particle in a periodic potential. [4]

***Q.6***

(a) Derive an expression for torque on a magnetic dipole inside a uniform magnetic field. Define magnetic moment. [7]

(b) Distinguish among dia, para and ferro magnetic materials on the basis of orientation of atomic magnetic moments under the external magnetic fields. [7]

***Q.7***

(a) State Poynting theorem and define Poynting vector. Use Maxwell's equations to derive the theorem. [10]

(b) Write down four Maxwell's equations in differential form. Name the terms. [4]

***Q.8***

(a) Write down Schrödinger's wave equation for a particle in one dimensional potential well given by: [10]

$$V(x) = 0, \quad 0 < x < a$$

$$= \infty, \quad x \leq 0 \text{ and } x \geq a$$

Solve it to obtain the energy eigen values.

(b) Derive Einstein's photo-electric equation. [4]

***Q.9 Write short note on any two of the following:*** **[7×2=14]**

(a) Resonance

(b) Polarization mechanisms in dielectrics

(c) Compton scattering

(d) Newton's ring experiment

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